MIE 100 – LEC 0103

Lecture 4

2.4 Rectangular Coordinates (x-y)

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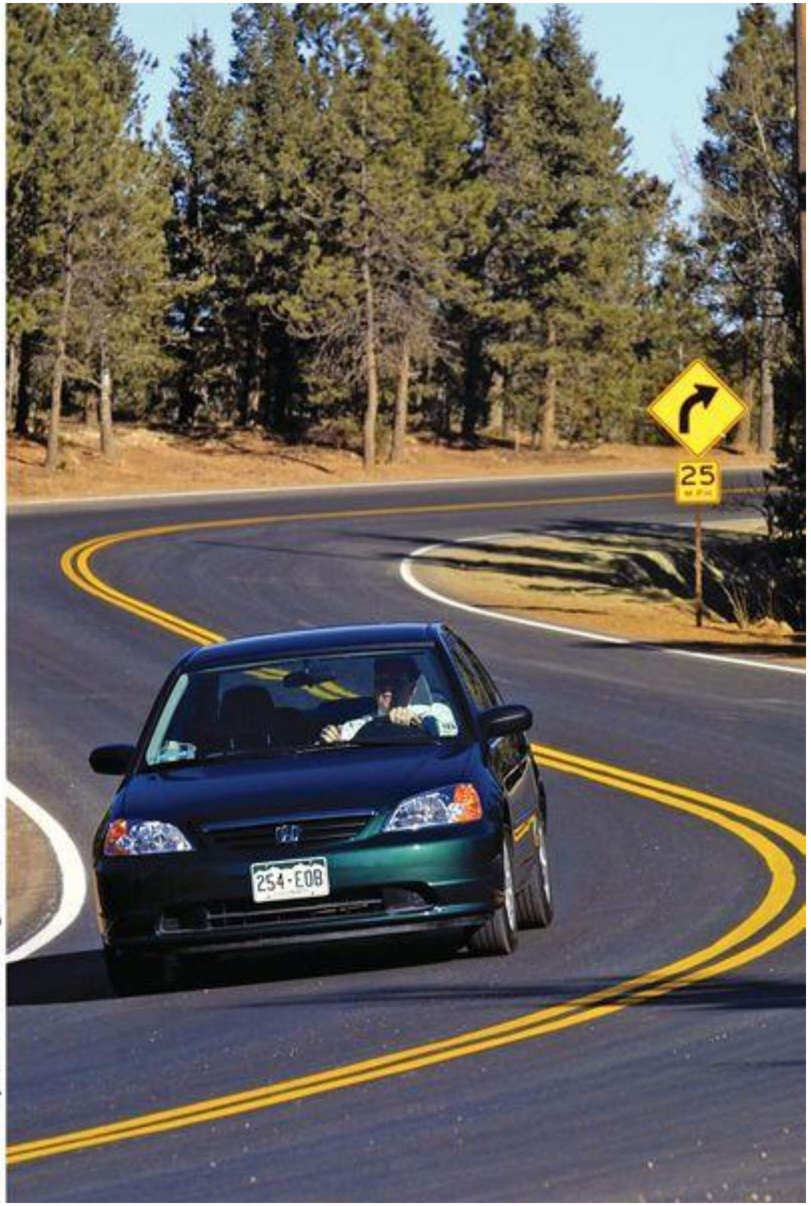


Sean Cayton/The Image Works

MIE 100 – LEC 0103

Lecture 4

Chapter 2: Kinematics of Particles

Howard Ho

- **2.1 Introduction**
- **2.2 Rectilinear Motion**
- **2.3 Plane Curvilinear Motion**
- **2.4 Rectangular Coordinates (x-y)**
- **2.5 Normal and Tangential Coordinates (n-t)**
- **2.6. Polar Coordinates (r-θ)**
- 2.8 Relative Motion (Translating Axes)
- 2.9 Constrained Motion of Connected Particles

The puzzle

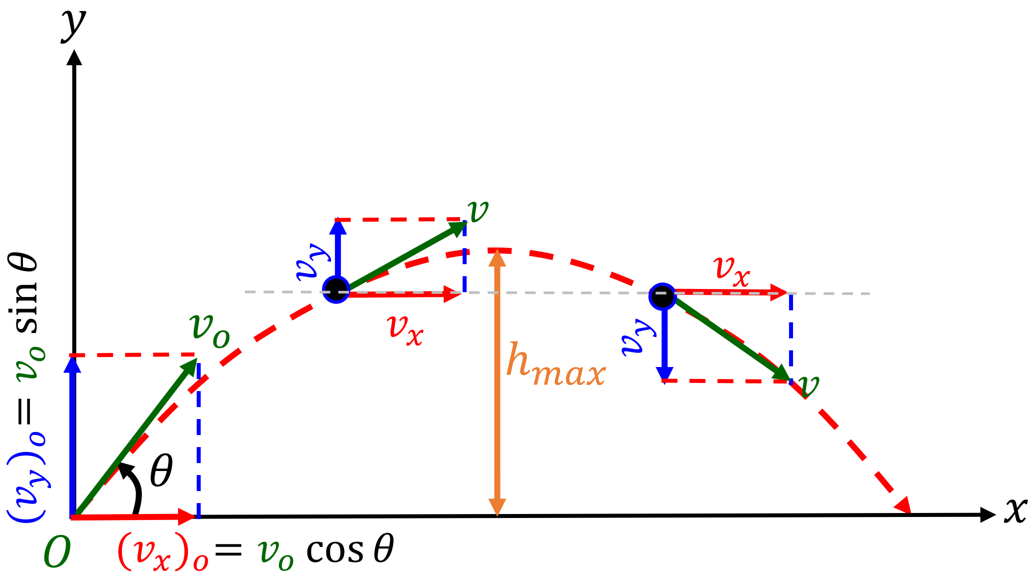
- You are tasked to place the landing mat for the human cannon stunt.

Learning Objective

In this lecture, you will learn to

- Apply the motion equations for analyzing projectile motion
- Analyze the free-flight motion of a projectile

Rectangular Coordinates: Projectile Motion



Assumptions
- Neglect aerodynamic drag
- Neglect curvature and rotation of earth
- Gravity is constant

- Projectile motion can be treated as two-dimensional rectilinear motions
- Horizontal direction
  - v_x is constant, therefore acceleration is zero

$$a_x = \frac{dv_x}{dt} = 0$$

- Vertical direction
  - Acceleration is constant (i.e., gravity):

$$a_y = \frac{dv_y}{dt} = a_c = -g$$

  - At maximum height, h_{max},

$$v_y = 0$$

Projectile Motion: Velocity and Position

Horizontal motion, $\vec{a}_x = 0$:

- Velocity

$$v_x = (v_x)_o$$

- Position

$$x = x_o + (v_x)_o t$$

Vertical motion, $\vec{a}_y = -g$:

- Velocity

$$v_y = (v_y)_o - gt$$

- Position

$$y = y_o + (v_y)_o t - 1/2(gt^2)$$

- Velocity-Position

$$v_y^2 = (v_y)_0^2 - 2g(y - y_o)$$

Recall: Constant Acceleration Eqns.

$$a_c = \frac{dV}{dt} = V\frac{dV}{ds}$$

Velocity:

$$V = V_o + a_c t$$

$$V^2 = V_o{}^2 + 2a_c(s - s_o)$$

Position:

$$s = s_o + V_o t + \frac{a_c t^2}{2}$$

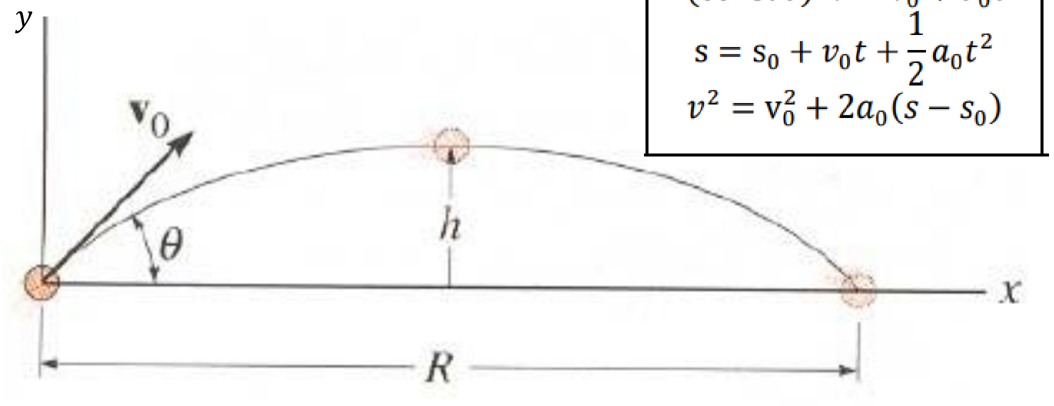
where s_o and V_o are the position and velocity at the initial time, t = 0

Example Problem [2.4-1]

Special cases	
$(a = 0)$	$s = s_0 + v_0 t$
(const a)	$v = v_0 + a_0 t$ $s = s_0 + v_0 t + \frac{1}{2} a_0 t^2$ $v^2 = v_0^2 + 2a_0(s - s_0)$

Given: v_o and θ

Find: The equation that defines y as a function of x



Assumptions:

Equations:

Trig rules:

$$tan\theta = \frac{sin\theta}{cos\theta}$$

$$\tan^2\theta + 1 = \sec^2\theta = 1/\cos^2\theta$$

Solutions:

Example Problem [2.4-2]

Given: Initial velocity v_o, launch angle θ

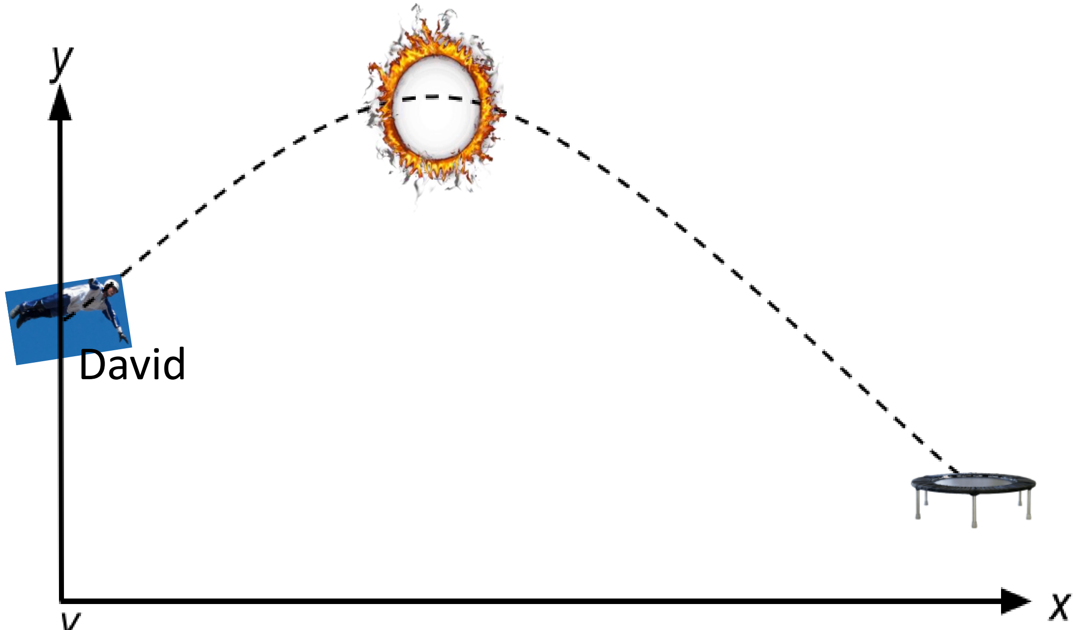
Find: **(i)** The horizontal location for placing the net

(ii) Vertical and horizontal location for ring of fire

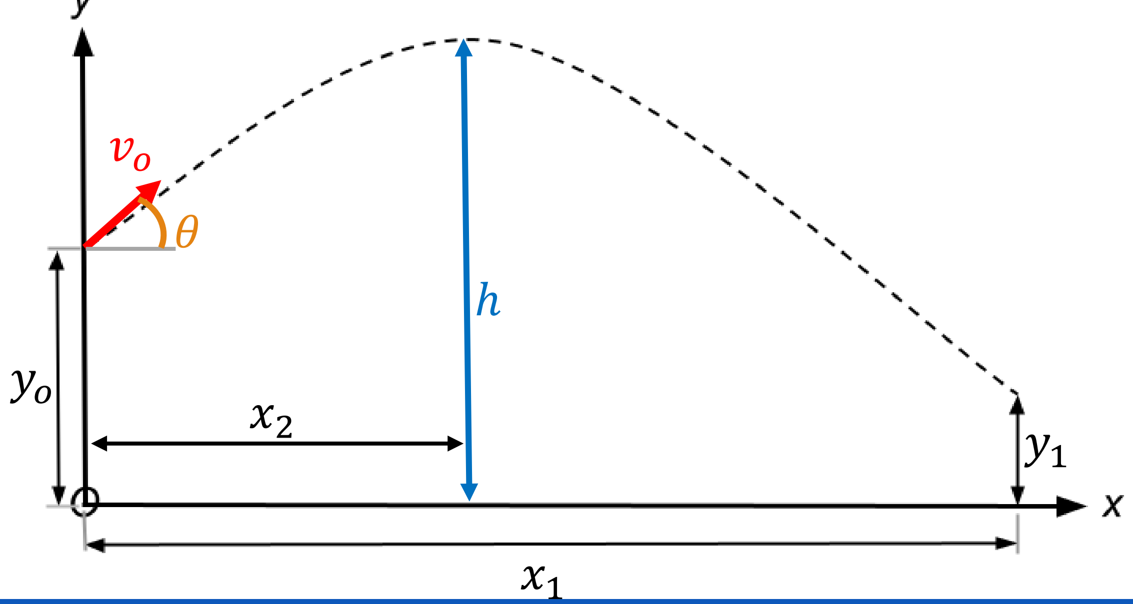
<u>Assumptions & Initial Conditions:</u>

$$y_o = 10m$$
$$y_1 = 5m$$
$$v_o = 10\frac{m}{s}$$
$$\theta = 30^o$$

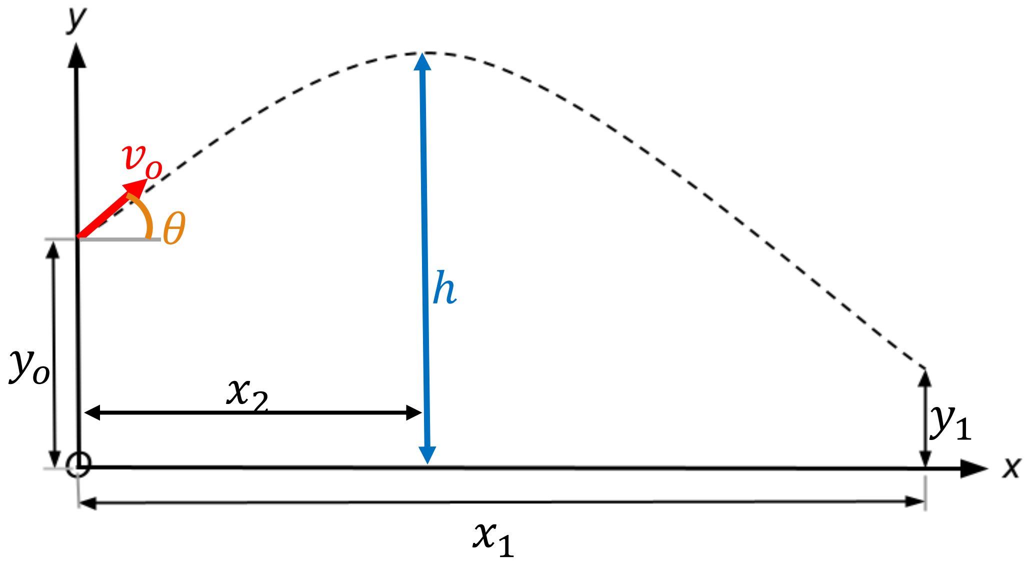
<u>Equations:</u>



Special cases	
$(\mathrm{a} = 0)$	$s = s_0 + v_0 t$
(const a)	$\mathrm{v} = \mathrm{v}_0 + \mathrm{a}_0 \mathrm{t}$ $\mathrm{s} = \mathrm{s}_0 + v_0 t + \frac{1}{2} a_0 t^2$ $v^2 = \mathrm{v}_0^2 + 2a_0(s - s_0)$



Solutions (i):



Special cases
(a = 0) $s = s_0 + v_0 t$
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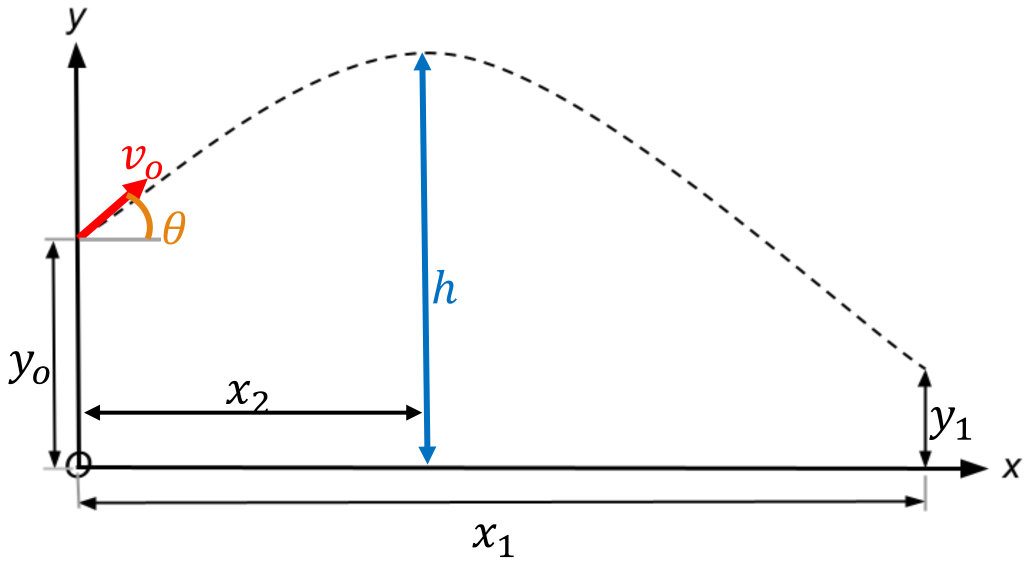
Solutions (ii):



Special cases
(a = 0) $s = s_0 + v_0 t$
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Special Cases: Horizontal Acceleration

Special cases	
$(a=0)$	$s = s_0 + v_0 t$
(const a)	$v = v_0 + a_0 t$ $s = s_0 + v_0 t + \frac{1}{2} a_0 t^2$ $v^2 = v_0^2 + 2a_0(s - s_0)$

Horizontal motion becomes, $\vec{a}_x = -a_x$:

- Velocity

$$v_x = (v_x)_o - a_x t$$

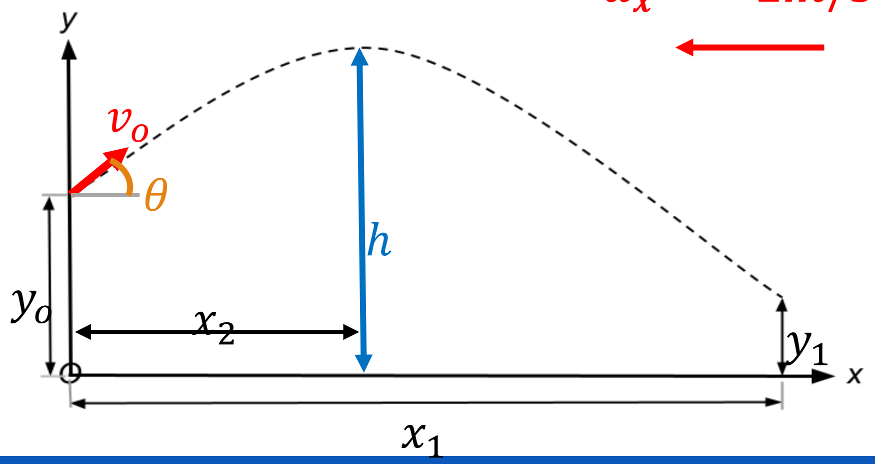
- Position

$$x = x_o + (v_x)_o t - 1/2(a_x t^2)$$

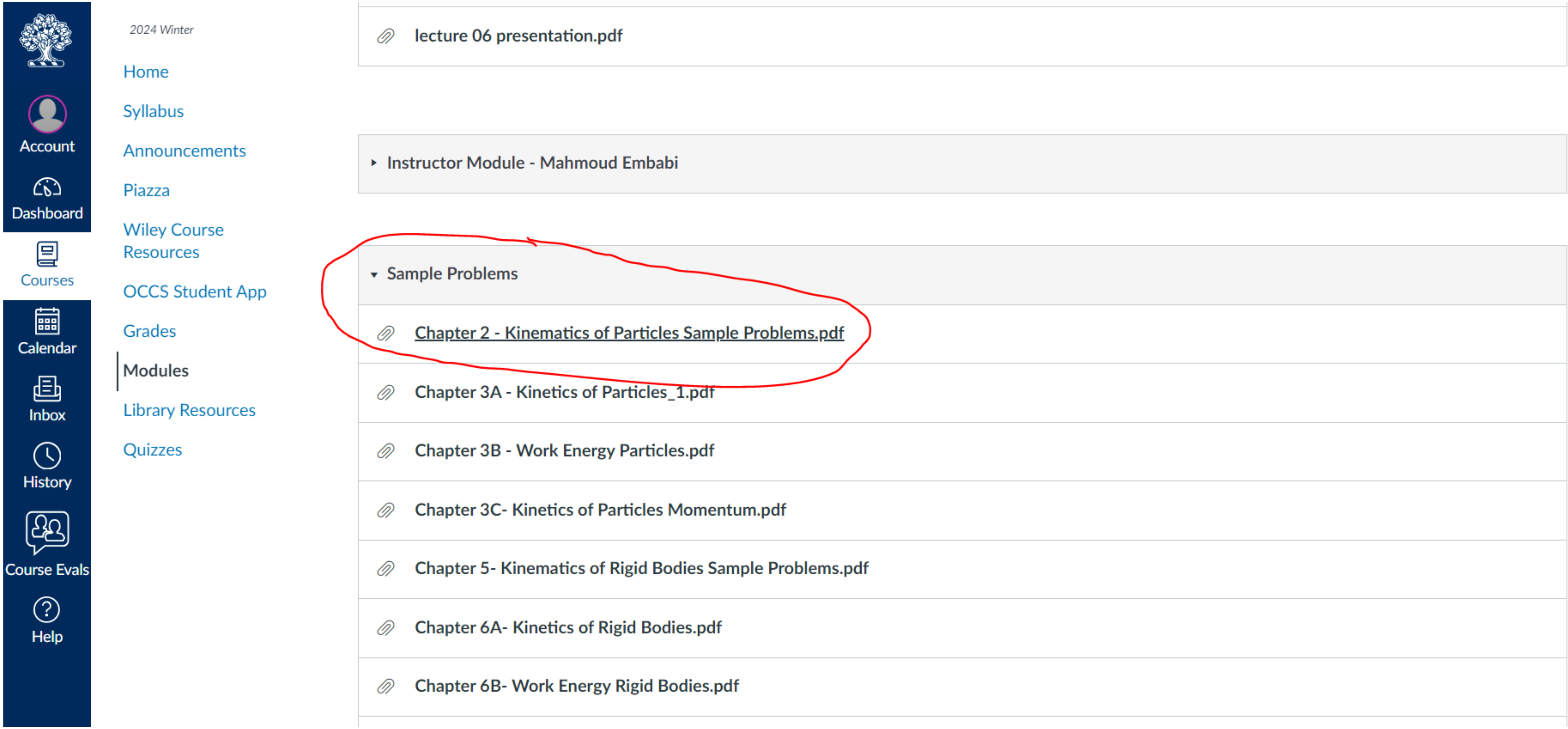
- Velocity-Position

$$v_x^2 = (v_x)_0^2 - 2a_x(x - x_o)$$

$a_x = -1m/s^2$

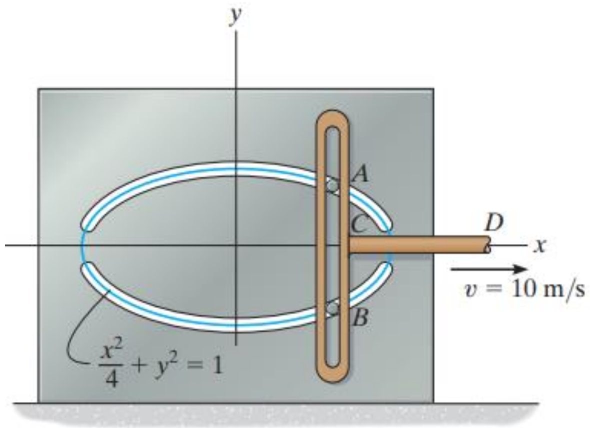


Practice Questions (from Quercus)

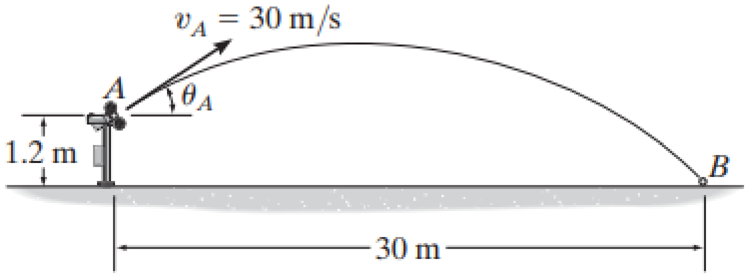


12–87.

Pegs A and B are restricted to move in the elliptical slots due to the motion of the slotted link. If the link moves with a constant speed of 10 m/s, determine the magnitude of the velocity and acceleration of peg A when $x = 1$ m.



12–91. The pitching machine is adjusted so that the baseball is launched with a speed of $v_A = 30$ m/s. If the ball strikes the ground at B, determine the two possible angles θ_A at which it was launched.



12–101.

It is observed that the skier leaves the ramp A at an angle $\theta_A = 25°$ with the horizontal. If he strikes the ground at B, determine his initial speed v_A and the speed at which he strikes the ground.

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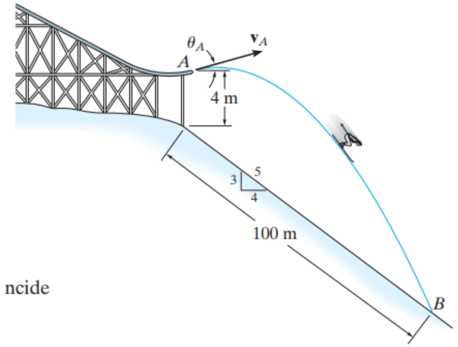

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Next lecture:
2.5 Normal and Tangential Coordinates (n-t)